



Research Funding and Higher Degree by Research Completions

AN ANALYSIS OF AUSTRALIAN HIGHER EDUCATION PROVIDERS, 2015-2024

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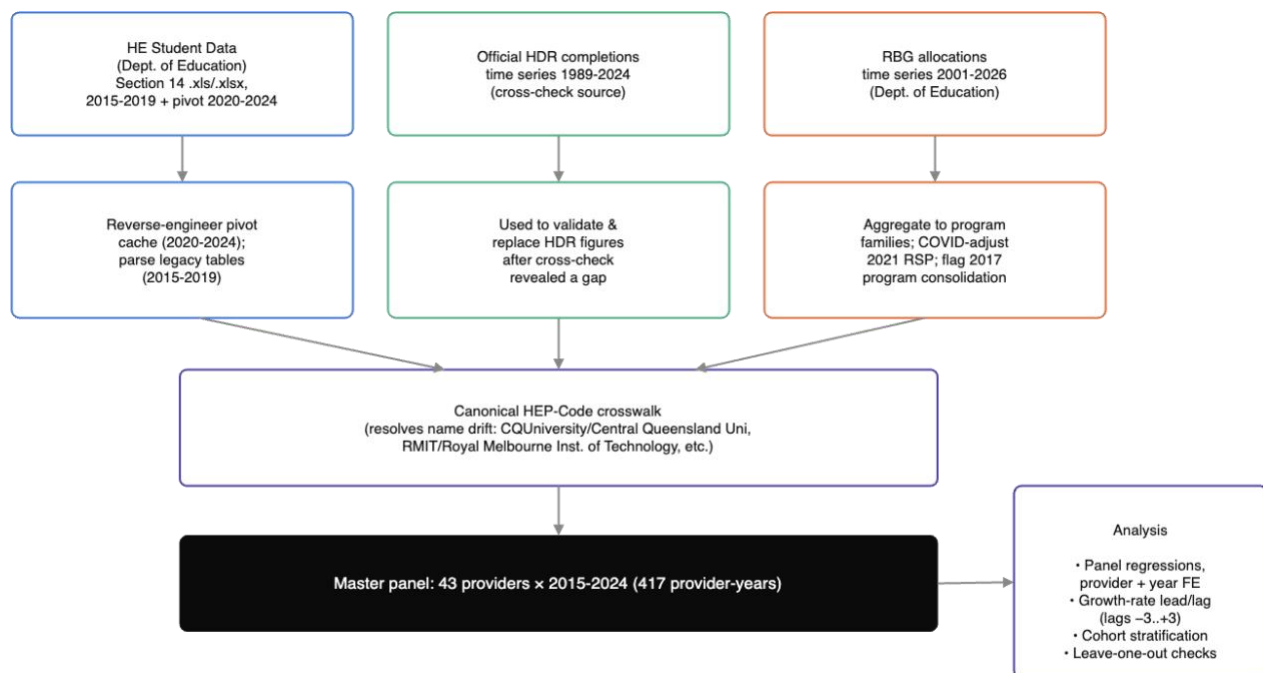
Section 1: Data and Model

1.1 Data sources

This analysis draws upon two primary public data series published by the Australian Government Department of Education, supplemented by an official historical time series used as an independent methodological validation source.

- **Higher Education Statistics** (Student Data collection): <https://www.education.gov.au/higher-education-statistics>, used for Higher Degree by Research (HDR) completions: Doctorate by Research + Master's by Research, by provider and year.
- **Research block grant (RBG) allocations time series**: <https://www.education.gov.au/research-block-grants/resources/research-block-grant-allocations-time-series>, used for research funding: the Research Training Program (RTP) and Research Support Program (RSP), by provider and year, 2001-2026.
- **Official HDR student completions time series (1989-2024)**: <https://www.education.gov.au/research-block-grants/resources/higher-degree-research-student-completions-time-series>, used as an independent cross-check of the completions figures reconstructed from the Student Data publications (Section 1.3).
- **RSP/RTP calculation methodology** and the **R&D income (HERDC) time series**, both published by the Dept. of Education under research-block-grants/resources, used in Section 2 Finding 2 to test whether the funding premium is mechanically explained by the legislated allocation formulas.

1.2 Pipeline



Grey arrows show data flow; the cross-check (aqua) fed back into the completions branch after a discrepancy was found and diagnosed (see Section 1.3).

Data preparation required structural reconstruction prior to panel integration. While the RBG allocation series was provided in a structured format, 2020-2024 HDR completion statistics were embedded within an Excel PivotTable, requiring extraction via pivot-cache XML parsing; 2015-2019 completions were compiled from five discrete annual statistical publications with inconsistent table numbering.

The 2020-2024 reconstruction recovered 98,800 individual pivot-cache records from the workbook's underlying XML, since the published file exposed only a single saved pivot view rather than the record-level

data beneath it. The reconstructed national HDR completions total (approximately 10,000-11,000 per year) was checked against the Department's own published summary figures before the panel was used for analysis.

1.3 Structural issues identified and resolved

- **Institutional name normalization.** Inconsistencies in institutional nomenclature across legacy files (e.g., CQUniversity vs. Central Queensland University; RMIT University vs. its RBG legal name, Royal Melbourne Institute of Technology; Curtin University vs. Curtin University of Technology) were resolved via a canonical Higher Education Provider (HEP) code crosswalk anchored to the RBG file's provider list.
- **2017 RBG program consolidation.** The 2017 reform consolidating six legacy programs (APA, IPRS, RTS, JRE, RIBG, SRE) into the RTP and RSP frameworks was harmonized by aggregating historical allocations into these two contemporary program families, so the series remains comparable before and after the reform.
- **2021 COVID-19 anomaly.** The introduction of a one-off supplementary pandemic funding allocation in 2021 (a documented \$1bn payment) created a structural outlier in RSP funding, alongside a temporary dip in national HDR completions plausibly reflecting submission extensions. Because both sides of the funding-completions relationship are affected in this single year, 2021 is excluded from the primary growth-rate specification, though retained for robustness testing.
- **Cell suppression.** Masked values ('<5' or 'np') in the legacy tables were imputed using documented interval midpoints and explicitly flagged per observation.
- **Suppression concentration.** Suppression was concentrated in the Master's by Research completions category: 34% of Master's by Research cells were suppressed in the 2015-2019 publications, compared with 8% for Doctorate by Research over the same years. Because Master's by Research is the smaller of the two HDR completion categories nationally, this concentration affects precision at small providers more than it affects the headline HDR total.
- **Methodological cross-check.** Reconciliation against the official HDR file (same HEP-Code scheme as the RBG file) revealed a high overall correlation ($r=0.99$), with material divergences concentrated in 2021 and a small number of other institution-years (e.g. Sydney -300, Melbourne -334). Diagnosis confirmed the crosswalk and overall completions counts were correct: the reconstructed panel's *total* (all-level) completions for these institution-years matched an independently published all-level series almost exactly, isolating the discrepancy to the Excel pivot tool's coarse course-level bucketing of HDR completions in COVID backlog-clearing years. The completions series was subsequently rebuilt directly from the official, purpose-built HDR file for the full analysis window, with the RBG-side cleaning and crosswalk unchanged.

1.4 Analytical approach

The final panel comprises **417 provider-years across 43 higher education providers (2015-2024)**, out of a theoretically balanced maximum of 430 (43 providers $\times$ 10 years). The panel is unbalanced by design, not by missing or suppressed data: the 13 absent provider-years reflect institutions without a full RBG allocation history across the window, Avondale University's first RBG allocation was in 2024 (having gained university status in 2021), and Torrens University Australia and Batchelor Institute of Indigenous Tertiary Education are each absent for 2015-2016. Relationships are estimated with log-linear OLS carrying provider and year fixed effects and standard errors clustered by provider, both in levels (funding on completions, testing for a cross-sectional concentration premium) and in year-on-year growth rates (testing lead/lag dynamics at lags -3 to +3, both directions and contemporaneously). Findings are stress-tested against: dropping 2021; splitting RTP vs. RSP; adding total (all-level) completions as an alternative scale control; stratifying by mission group (Go8, ATN, IRU, RUN, non-aligned); and a leave-one-Go8-university-out check.

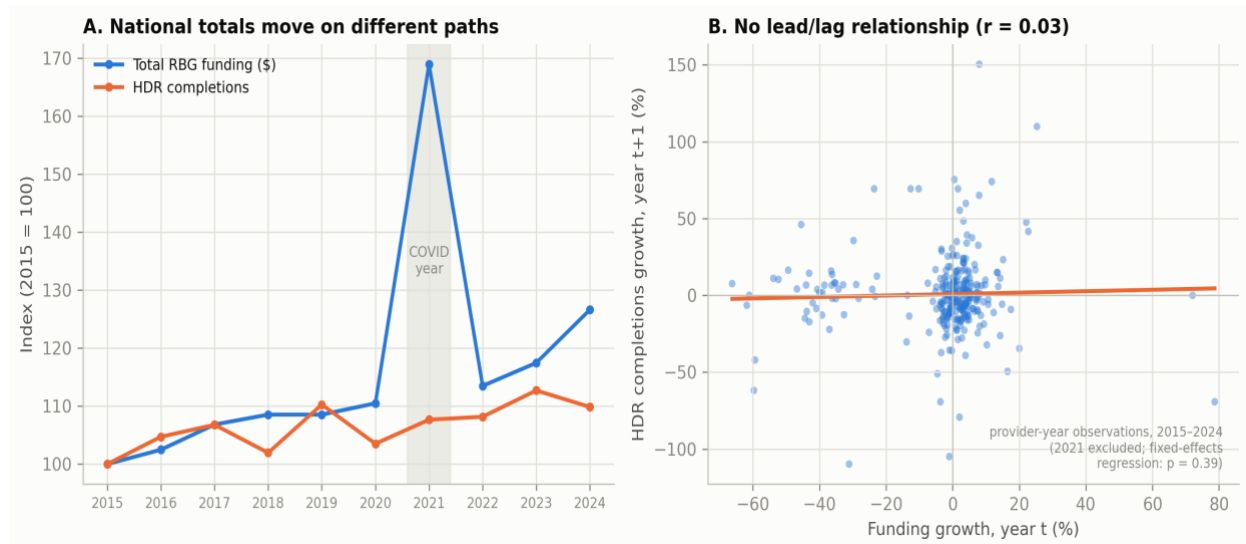
1.5 Limitations

- **Window length.** Ten annual observations per provider restrict the capacity to detect multi-year cyclical dynamics; a relationship operating on a 3-5-year cycle, plausible given RTP's own multi-year formula, may not be visible in year-on-year growth rates even if it exists.
- **Correlational, not causal.** Fixed-effects specifications mitigate time-invariant unobserved heterogeneity and common shocks by year, but do not establish causal identification; an unobserved, time-varying provider-specific trend correlated with both funding and completions growth remains a potential source of confounding.
- **Small-provider suppression.** Cell-count masking in the legacy tables is concentrated among the smallest providers, which contribute little to the funding totals driving either finding, though their individual growth rates are noisier as a result.
- **Forward changes not in scope.** The 2026 merger of the University of Adelaide and University of South Australia into "Adelaide University" falls outside the 2015-2024 window and does not affect either finding, though it would require a crosswalk update before the panel is extended forward.

Section 2: Key Findings

FINDING 1

No statistically significant funding-completions dynamic detected once institution size is controlled for



Source: Dept. of Education RBG allocations time series; HDR completions time series (1989-2024). $n=417$ provider-years, 2015-2024 ($n=850$, 2001-2024 for the extended-window check).

Raw provider-level correlation between RBG funding and HDR completions is high ($r \approx 0.91$), a figure that on its own might suggest a straightforward funding-drives-completions relationship. This correlation does not withstand closer scrutiny, however: it is measured in levels, where larger, more established institutions mechanically exhibit greater volumes across every academic input and output, more funding, more completions, more staff, more students. Bivariate levels of this kind do not identify whether a change in one variable precedes a change in the other.

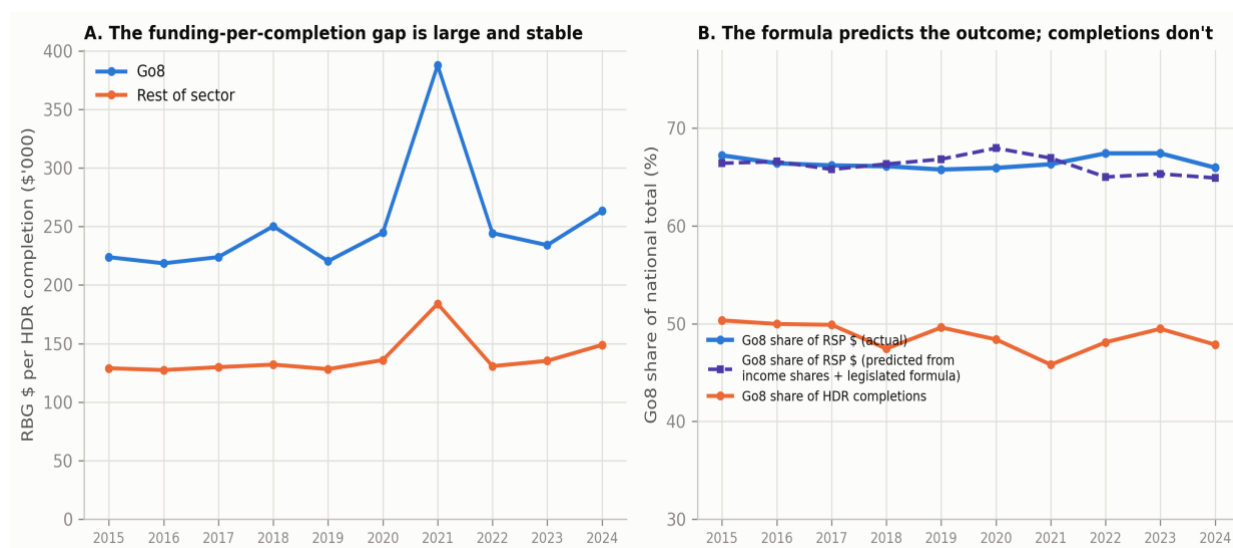
Testing the underlying dynamic, whether year-on-year funding growth leads, lags, or moves contemporaneously with completions growth, removes the scale effect via provider and year fixed effects and tests lags -3 to $+3$ in both directions, as well as the same-year relationship. Essentially none of the resulting estimates are statistically significant: funding growth $\rightarrow$ next-year completions growth (coefficient = -0.33 , $p = 0.39$); completions growth $\rightarrow$ next-year funding growth (coefficient ≈ 0.00 , $p = 0.96$); same-year ($p = 0.14$). This result holds using Total RBG or RTP alone, with and without 2021, and holds even more strongly on an extended 2001-2024 window ($n = 850$ vs. 290 , $p = 0.29$ - 0.71 across variants). One isolated exception is noted: the same-year relationship becomes marginally significant ($p = 0.03$, negative coefficient) specifically when 2024 is excluded, a result not replicated in the lead specification or the extended window, and consistent with chance given the number of specifications tested.

Panel A illustrates why. Indexed to 2015, RBG funding climbs steadily and spikes sharply during the 2021 COVID top-up, while HDR completions follows a materially different, choppy path; even a 69% one-year funding increase produced no corresponding movement in completions. Panel B shows the same result at the provider level: funding growth and next-year completions growth form a flat, uncorrelated cloud ($r = 0.03$).

Interpretation: RTP's own allocation formula is explicitly a multi-year rolling average, not a same-year or one-year-lag mechanism; this result is consistent with that design rather than a contradiction of it. This has a direct implication for KPI design: annual funding-to-output tracking should not be expected to move in lockstep, and the absence of same-year movement should not be read as a sign of underperformance.

FINDING 2

Go8's funding premium per completion is not about completions; it's the legislated formula's income weighting



Source: Dept. of Education RBG allocations time series; HDR completions time series (1989-2024); R&D income (HERDC) time series; RSP/RTP calculation guidance. $n=417$ provider-years, 2015-2024.

Group of Eight (Go8) universities, 8 of the 43 providers in the panel hold roughly 48% of the sector's HDR completions but 62-64% of RBG dollars, a gap that has held steady across all ten years and appears independently at every single Go8 university (1.56x-2.24x the non-Go8 \$-per-completion average; not 2-3 outliers carrying the result).

This disparity is driven by the structural parameters of the funding formulas established under the Other Grants Guidelines (Research) 2017 and the Higher Education Support (Commonwealth Scholarships) Guidelines 2025, which show **RSP is 47% competitive R&D income share + 53% engagement income share, with zero HDR completions weighting**, while RTP is 25% + 25% income plus **50% weighted HDR completions**. Go8 institutions secure approximately 68.5% of national competitive research income and 64.2% of engagement income, substantially exceeding their 48% completions share. Reconstructing Go8's predicted RSP share using nothing but these income shares and the legislated weights reproduces its *actual* observed RSP share to within 1-2 percentage points every year (Panel B): the funding pattern reflects the formula operating as designed, not an underlying completions story.

Consistent with this, the regression premium is **+124% in RSP** ($p < 0.0001$) but only **+21% in RTP** ($p = 0.10$, not significant) once HDR completions volume is controlled for, precisely because RTP's formula is half-diluted by a completions component RSP does not have. Using total, rather than HDR-specific, completions as the control renders RTP significant as well; the choice of denominator matters, and HDR completions is the measure RTP's own formula actually uses. Other mission groups (ATN, IRU, RUN) show no significant premium over non-aligned providers.

Interpretation: Consequently, the observed funding premium is consistent with the statutory design of the allocation formulas rather than an output-efficiency premium. Attributing block grant concentration to HDR completion volume alone misidentifies the underlying income-weighted allocation mechanism; Go8's advantage is concentrated in R&D income (competitive grants and industry engagement), a materially different lever than HDR training capacity.

Appendix

The full data pipeline, cleaning scripts, regression code, and chart-generation code for this report are published in a documented Git repository, including the reconciliation crosswalk, the pivot-cache reconstruction, and every robustness check referenced in Section 1.4. The repository also includes a three-page Power BI dashboard extending both findings with interactive slicers and provider-level detail (screenshots in the repository's README).

Git repository: <https://github.com/mithizs08/UTS-Research-KPI-Analysis>

Note on reproducibility: all analyses and visualisations in this report were generated programmatically from the source files using Python and Power BI; no manual data entry was used. Suppressed-cell imputation and the 2021 COVID exclusion are documented, auditable choices rather than silent adjustments, described in Section 1.3 and implemented as explicit, labelled steps in the repository code.